

Phase 3 Activity Report

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Activity Overview

During Phase 3, we significantly refined our pipeline developed in Phase 2, aiming for a more generalizable, robust, and fair predictive model for cognitive function scores in older adults. Several methodological shifts were made to simplify the pipeline, improve computational efficiency, expand applicability across datasets, and enhance interpretability for potential clinical use.

Key changes from Phase 2 to Phase 3:

- **Expanded dataset coverage:** Added three new countries (India, Mexico, UK) to the modeling pipeline alongside the US, enabling cross-country analysis and improved model generalizability.
- **Simplified target definition:** Removed prediction of 2016-2021 cognitive score differences and shifted to predicting absolute cognitive scores standardized as z-scores. This makes the model applicable even in settings lacking longitudinal cognitive measures.
- **Efficiency improvements:** Eliminated multiple-seed averaging to reduce computational load without significant performance loss.
- **Enhanced explainability:** Improved SHAP-based interpretation by statistically regressing SHAP values on feature values, providing global interpretability rather than relying only on local explanations.
- **Uncertainty estimation:** Integrated Jackknife+ for computing individual-level prediction intervals, adding clinically useful uncertainty measures.
- **Fairness integration:** Moved fairness adjustments inside individual models (via inverse-frequency weighting) instead of relying on external blending mechanisms.
- **Pipeline reorganization:** Rewrote code to follow a fit-predict pattern, reducing redundant steps and improving maintainability.
- **Model pruning:** Removed Lasso and TabPFN from the pipeline due to underperformance on larger, multi-country data and minimal gains in predictive accuracy, respectively.
- **Literature integration:** Linked feature importance findings directly to existing scientific literature to strengthen interpretation and potential clinical implications.

Data

Additional datasets integrated in Phase 3:

- India (LASI + LASI-DAD)
- Mexico (MHAS + Mex-Cog)
- United Kingdom (ELSA + ELSA-HCAP)

Impact of adding new countries:

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- Added heterogeneity in socio-demographic and clinical features, allowing for broader generalizability.
- Enabled testing cross-country transfer learning. Training on pooled data from multiple countries generally reduced RMSE compared to training on single-country data (except in Mexico).
- Model performance remained acceptable even when tested on a completely unseen country, suggesting good potential for transfer learning.

These enhancements addressed a limitation from Phase 2, where the model focused solely on US data, potentially constraining its global utility.

Generalizability

Key efforts to enhance generalizability and fairness:

- Cross-country transfer tests: Systematically trained models on subsets of countries and tested on held-out countries to assess how well the model generalized to new populations. Shared patterns improved performance in most countries except Mexico, indicating cultural or contextual differences not fully captured by harmonized features.
- Wave exclusion tests: Tested robustness to missing recent data by dropping the most recent survey waves and evaluating performance degradation. RMSE increased modestly, indicating the model remains usable even with older data, an important feature for longitudinal research where newer waves may not be available.
- Data reduction robustness: Randomly removed up to 50-70% of features or observations. RMSE stayed largely stable, suggesting resilience against incomplete or missing data, which is frequent in real-world clinical datasets.
- Bias mitigation: Applied fairness constraints via inverse-frequency weighting in model training. Fairness Index (FI) improved across country, gender, and education groups after correction.

These steps align with recommendations in the broader machine learning literature to evaluate subgroup performance systematically (Obermeyer et al., 2019; Rajkomar et al., 2018).

Methodology

New methodological experiments in Phase 3:

- Jackknife+ for Prediction Intervals: Implemented Jackknife+ (Barber et al., 2020) to quantify uncertainty around individual predictions. This provides clinicians with 95% confidence intervals, crucial for informed decision-making, especially in high-risk domains like dementia prediction.
- SHAP Regression for global interpretability: Beyond calculating local SHAP values, we regressed SHAP values against feature values to produce global coefficients and confidence intervals. This transforms local explanations into population-level insights, as recommended by Molnar (2020).
- Model simplification: Removed sequential prediction steps and multiple-seed averaging from Phase 2. The ensemble of LightGBM, CatBoost, and XGBoost already captured most of the predictive signal without needing sequential stacking or repeated seeds.
- Fairness embedded into base models: Instead of post-model adjustments, fairness was incorporated into individual model training, leading to smoother integration and less performance penalty than in Phase 2.

- Feature selection and engineering: Tested the utility of logarithmic transformations and group-level aggregate features from Phase 2. Found only modest incremental gains over basic features, prompting us to prioritise simplicity and interpretability.
- Removed:
 - TabPFN - despite earlier Phase 2 success, added no performance boost on larger datasets.
 - Lasso Regression - performed substantially worse on the new multi-country dataset.

These methodological adjustments reflect a deliberate trade-off toward simpler, faster, and more deployable solutions without sacrificing predictive performance.

Reference

Key references shaping Phase 3 decisions:

- Gross et al. (2023) - Described the harmonized cognitive function scores used as the Phase 3 target variable. Provided the methodological basis for cross-country score standardization.
- Molnar (2020) - Advocated for moving from local to global interpretability, motivating our SHAP regression approach.
- Barber et al. (2020) - Introduced Jackknife+, which we used for individual-level uncertainty quantification.
- Obermeyer et al. (2019); Rajkomar et al. (2018) - Highlighted the necessity of fairness analysis in healthcare ML applications.
- Kuiper et al. (2015); Saenz et al. (2022); Hudomiet et al. (2024) - Provided evidence linking social determinants such as rural residence, loneliness, and discrimination to cognitive decline, influencing our feature interpretation.
- Livingston et al. (2020); Stern (2012) - Emphasized the protective role of education in cognitive health, directly supporting our findings that education is the strongest positive predictor.

Our work integrated these insights to align modeling choices with established clinical and epidemiological understanding of cognitive aging.

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